

AYANTIKA NANDI

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EDUCATION

Indiana University Bloomington

Master of Science, Data Science

Aug 2023 – May 2025

Bloomington, Indiana

Sikkim Manipal Institute of Technology

Bachelor of Technology, Computer Science and Engineering

Aug 2017 – May 2021

Sikkim, India

SKILLS

- **Core Languages:** Python, SQL, Java, C/C++, JavaScript/TypeScript
- **Data Engineering:** PySpark, Dataflow, Dataproc, Kafka, dbt, Hive
- **Cloud Platforms & Databases:** GCP, AWS, Snowflake, PostgreSQL, MySQL
- **Data Modeling & Warehousing:** Star/Snowflake Schema, Slowly Changing Dimensions (SCD), performance tuning
- **Orchestration & MLOps:** Apache Airflow (modular DAGs), MLflow, Docker, Kubernetes, Databricks
- **Machine Learning & Tools:** TensorFlow, PyTorch, scikit-learn, OpenCV; REST API

Certifications: AWS AI Practitioner [🔗](#), Neo4j Certified Pro [🔗](#), LLM Engineer [🔗](#)

WORK EXPERIENCE

Indiana University Bloomington

Software Engineer

Bloomington, Indiana

May 2024 – Present

- Engineered modular backend services using Python, PostgreSQL, and AWS EC2, enabling scalable processing of 10K+ legal documents and reducing manual review time by 35%
- Designed and deployed a computer vision pipeline using OpenCV, integrated with AWS S3 for secure document storage and retrieval, increasing redaction detection accuracy from 65% to 83%
- Implemented a multilingual NLP classification pipeline with spaCy and AWS Lambda, improving document processing speed by 40% while maintaining a serverless architecture
- Integrated automated testing with PyTest and continuous deployment via GitHub Actions, achieving 90% test coverage and ensuring robust, reliable production releases
- Developed Python scripts to clean and standardize case data, reducing data inconsistency by 35%

Cognizant Technology Solutions

Data Engineer - Supply Chain

Hyderabad, India

Sept 2021 – April 2023

- Built and optimized 50+ ETL/ELT pipelines on GCP (Cloud Storage, BigQuery, Dataflow, Pub/Sub, Composer), enabling reliable data ingestion at scale and reducing pipeline failures by 35%
- Processed real-time and batch data streams with Cloud Dataflow (Python), integrating Pub/Sub to BigQuery and cutting event-to-dashboard latency by 45%
- Orchestrated workflows with Apache Airflow using GCP operators, modular DAGs, and dynamic task generation, reducing manual interventions by 60% and improving scheduling efficiency
- Ensured data quality by implementing dbt schema tests, custom validations, and automated BigQuery checks, raising data accuracy to 99%+ across downstream reporting
- Optimized queries and storage by applying partitioning, clustering, and SQL tuning in BigQuery, improving query performance by 30% and lowering costs by 15%

Data Engineer intern

April 2021 – Sept 2021

- Developed scalable ETL pipelines using Apache Spark and Hive to process over 2TB/day, increasing reporting system performance by 25%
- Created reusable PySpark modules for transformation and validation, adopted by the team to reduce ETL job runtimes by 20% and standardize data pipelines

PROJECTS

HemoTrack- Cloud-based Blood Bank Management System | React.js, Node.js, AWS, JavaScript

Feb 2025- Mar 2025

- Designed a cloud-based real-time blood inventory system, with a donor-recipient matchmaking efficiency of 97%
- Implemented secure and scalable cloud infrastructure using AWS, improving data accessibility by 20%

A/B Testing on Personalized Recommendations | Python, SQL, SciPy, Tableau

Dec 2024- Jan 2025

- Conducted simulated A/B testing on the UCI Online Retail dataset using Python + SciPy, modeling the impact of personalized banners
- Achieved a 13.4% uplift in projected conversions ($p < 0.01$) and visualized actionable insights in Tableau, showcasing data-driven product experimentation